Article

Verified Localized Episodic Grounding for Auditable LLM-Assisted Network Traffic Analysis: Counterexample Diagnosis and Gating

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Abstract: Memory-augmented large language models (LLMs) can assist network intrusion triage by retrieving historical flows without parameter updates, yet compressing raw telemetry into natural-language rules creates a fundamental tension between single-record faithfulness and cross-flow specificity. We present Verified Localized Episodic Grounding (VLEG), an auditable memory-consolidation and hybrid-inference framework that verifies candidate reflections against their labeled source records within a bounded refinement budget K and pairs admitted rules with raw flow evidence. In a 16-scene symbolic sanity check at 7B and 70B scales, bounded verification eliminates source-label inversions (100% observed source-label agreement on admitted episodes versus 41.7%–50.0% for unverified proposals), reaching accuracy comparable to raw in-context retrieval at 70B (91.67% versus 89.58%, 44/48 versus 43/48) while trailing raw retrieval at 7B (75.0% versus 87.5%). On a 32-episode few-shot benchmark over 1000 deduplicated UNSW-NB15 test fingerprints across five memory seeds, unverified single-step reflection mislabels 95.0% (76/80) of malicious historical flows as benign, causing attack recall to collapse to 6.17% (42.66% Macro-F1). Standard VLEG ($K = 3$) restores 100% source-label agreement (158/158) and raises attack recall to 77.74% and Macro-F1 to 59.68% (+17.02 percentage points, pp), but trails zero-cost raw flow retrieval (63.92% Macro-F1) by 4.24 pp due to a 48.78% false positive rate (FPR). Using a deterministic counterexample compiler that parses 99.4% (157/158) of admitted rules into executable Boolean predicates, we show that 72.0% (113/157) of source-consistent rules match contrary-label historical flows. Integrating this compiler as a within-seed counterexample gate (VLEG-CG) without additional LLM calls reduces the FPR by 9.35–10.92 pp (down to 37.86%–39.43%, comparable to 32-flow Random Forest at 40.66% and GBDT at 39.22%) and recovers Macro-F1 to 63.32%–63.55%.

Keywords: episodic memory; hallucination mitigation; large language models; network traffic analysis; intrusion detection; rule over-generalization; verified memory consolidation

1. Introduction

1.1. Background and Motivation

Large language models (LLMs) can use in-context demonstrations to support classification and incident analysis without parameter updates [1, 2]. In network traffic analysis and Security Operations Center (SOC) intrusion triage [3–6], security teams increasingly explore LLM agents to link novel flow telemetry to historical incidents via human-auditable verbal rules. Whereas traditional supervised classifiers such as Random Forest (RF) and Gradient Boosting Decision Trees (GBDT) excel at high-throughput binary filtering when trained on tens of thousands of labeled flows, cold-start triage often operates with only a small curated bank of investigated episodes where analysts require explicit rule rationales. However, finite and imperfectly utilized context windows make the selection and verification of this historical evidence critical [7, 8].

Persistent-memory architectures extend an agent's access to experience beyond the current prompt [9–13]. Retaining raw flow records preserves instance-level detail, whereas converting records into natural-language reflections or global summaries supplies compact decision heuristics. Yet rule compression introduces two distinct failure modes: (1) unverified reflections generated without label supervision can hallucinate class labels,

and (2) locally verified reflections that accurately describe a single source flow can still assert overly broad conditions that trigger false alarms across benign background traffic.

Crucially, single-episode source fidelity and cross-episode specificity are distinct requirements. Checking a candidate explanation against its originating record alone can catch label inversions or fabricated attributes, but cannot determine whether the selected feature thresholds separate classes across the broader traffic distribution. Evaluating reflective memory systems therefore requires auditing admission coverage, source-label agreement, cross-episode counterexample rates, and downstream classification trade-offs together.

1.2. Research Positioning and Study Scope

Memory-augmented language models retain and reuse experience through episodic retrieval, local reflection, and global semantic consolidation [14,15]. Feedback-driven agents revise behavior using trial rewards [16,17], while global hypothesis frameworks synthesize and verify rules across multiple records [18,19]. Rather than claiming that LLM reflection outperforms supervised tabular classifiers or raw retrieval in unconditional accuracy, this study conducts a rigorous diagnostic audit of per-episode verification, rule over-generalization, and counterexample gating under strict few-shot memory budgets.

Specifically, we address three engineering questions:

1. Does checking each local reflection against its labeled source episode before memory admission prevent the label-inversion and recall collapse of unverified reflection?
2. Why do 100% source-consistent local rules still elevate false alarms when paired with raw telemetry on deduplicated network flows?
3. Can a deterministic counterexample compiler screen out over-generalized local rules within the historical memory bank to suppress false positives without extra LLM calls?

1.3. Proposed VLEG Framework and Counterexample Gate

We propose Verified Localized Episodic Grounding (VLEG), an inspectable memory-consolidation and hybrid-inference framework paired with a deterministic counterexample compiler and gate (VLEG-CG). During consolidation, a generator proposes a natural-language reflection from label-free episode attributes, and a separate verifier checks the candidate against the labeled source episode. Rejected candidates receive diagnostic critique and are revised within a maximum budget K, admitting the first accepted reflection alongside its source record.

To resolve the precision degradation caused by single-episode verification, our deterministic counterexample compiler parses admitted natural-language traffic rules into executable Boolean predicates over flow features. By evaluating each compiled rule against the remaining historical episodes within the same memory seed, VLEG-CG identifies and gates out over-generalized rules that match contrary-label historical flows, falling back to raw episodic evidence when retrieved rules fail cross-episode specificity.

1.4. Summary of Contributions

We evaluate VLEG through a 16-scene symbolic sanity check (at 7B and 70B scales) and a 25,000-prediction public network intrusion benchmark on deduplicated UNSW-NB15 flows [37], contextualized against both few-shot and full-dataset Random Forest and GBDT baselines. Our primary contributions are four-fold:

- **Auditable Local Consolidation and Counterexample Formulation:** We formulate VLEG and explicitly characterize the theoretical boundary between single-episode source verification $\mathcal{V}_\phi((x_i, y_i), r_i^*)$ and cross-episode counterexample specificity $C(r_i^* \mid \mathcal{E}_{-i})$.
- **Controlled Symbolic Sanity Check:** Using a 16-scene synthetic environment with known ground-truth rules and non-causal distractors, we verify that bounded refinement ($K = 3$) eliminates source-label inversions (100% observed source-label agreement versus 41.7%–50.0% for unverified proposals) and recovers admission coverage (93.8%–100%), while confirming that single-episode checks cannot filter out coincidental distractor attributes.
- **Mechanistic Diagnosis of Reflection Collapse and False Alarms on UNSW-NB15:** Across five 32-flow memory seeds on 1000 shared UNSW-NB15 test fingerprints, our deterministic compiler (99.4% rule

parseability) reveals that unverified single-step reflection mislabels 95.0% (76/80) of malicious historical flows as benign, causing test attack recall to collapse to 6.17%. Standard VLEG restores 100% source-label agreement (158/158) and lifts attack recall to 77.74% (59.68% Macro-F1), but trails raw retrieval (63.92% Macro-F1) by 4.24 pp because 72.0% (113/157) of source-consistent rules match contrary-label historical flows.

- **Counterexample-Gated VLEG (VLEG-CG) and Tabular ML Benchmarking:** We demonstrate that gating over-generalized rules via within-seed counterexample screening (VLEG-CG) slashes VLEG's false positive rate by 9.35–10.92 pp (from 48.78% down to 37.86%–39.43%) and recovers Macro-F1 to 63.32%–63.55% at zero additional LLM cost. We further benchmark few-shot (32-flow) and full-train (88, 699-flow) Random Forest and GBDT models on the exact same deduplicated test split to establish clear operational boundaries for SOC deployment.

2. Related Work

2.1. Episodic Memory and Demonstration Retrieval

In-context learning adapts LLMs to classification tasks using prompt demonstrations without gradient updates [1], while retrieval-augmented generation (RAG) conditions predictions on external evidence retrieved by embedding similarity [2]. LongMem [10], MemGPT [9], MemoryLLM [11], and HippoRAG [12] extend context capacity via persistent memory banks. Related studies examine demonstration selection [20, 21] and long-context utilization limits [7, 8, 13–15].

In network traffic analysis, raw episodic retrieval provides a strong, zero-consolidation-cost baseline: each retrieved flow retains its exact feature values and ground-truth label, preserving local exceptions that compressed summaries discard. However, raw retrieval supplies no explicit verbal rule for analyst verification. In our public-flow evaluation, we share label-free feature retrieval across all local memory conditions so that classification differences strictly reflect memory representation and gating rather than candidate-selection variance.

2.2. Local Reflection and Self-Verification

Agent memory frameworks frequently convert raw interactions into verbal reflections. MemoryBank updates interaction summaries over time [22], Generative Agents synthesize higher-level reflections from event streams [23], and Reflexion converts binary task feedback into verbal critiques for subsequent trials [16]. Related works investigate procedural skill libraries [17], structured agentic memory [24], timeline memory [25], and self-critique or hallucination verification [33–36].

A fundamental distinction separates outcome feedback from source-episode admission checking. Outcome feedback signals whether a downstream prediction succeeded, whereas a source admission check evaluates whether a candidate explanation agrees with the historical record from which it was induced. As we show both theoretically and empirically, single-episode verification eliminates label hallucinations but cannot prevent rule over-generalization unless paired with cross-episode counterexample screening.

2.3. Semantic Consolidation and Intrusion Detection Systems

Global semantic consolidation compresses multiple episodes into unified hypotheses [18, 19, 26–32]. In network intrusion detection, pre-trained packet transformers such as ET-BERT [3] and supervised tree ensembles (e.g., Random Forest and GBDT/XGBoost on UNSW-NB15 [37] and CICIDS2017 [4]) provide strong discriminative performance when trained on large labeled datasets. Recently, Houssel et al. [5] evaluated LLMs for explainable intrusion detection, and Islam et al. [6] introduced MA-IDS, a multi-agent RAG framework with an experience library for IoT traffic.

Our study complements prior LLM-IDS research [5,6] in two ways. First, we evaluate traditional tree ensembles (RF and GBDT) alongside LLM memory strategies on the exact same deduplicated UNSW-NB15 splits under both 32-flow few-shot and 88,699-flow full-train regimes. Second, we isolate why unscreened natural-language traffic rules inflate false alarms and demonstrate how deterministic counterexample gating restores precision.

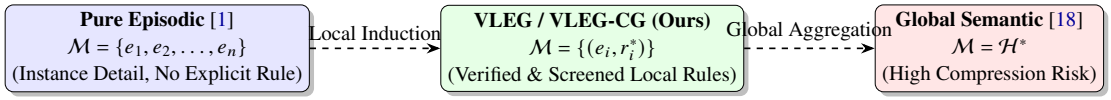


Figure 1. Positioning of VLEG across the memory abstraction spectrum. VLEG pairs raw episodic records with source-verified and counterexample-screened local abstractions.

3. Proposed Methodology

3.1. Problem Formulation and Memory Spectrum

Let $\mathcal{E} = \{e_1, \ldots, e_n\}$ denote a support bank of labeled historical episodes available during memory consolidation, where each episode $e_i = (x_i, y_i)$ comprises observed feature attributes $x_i \in \mathcal{X}$ and a ground-truth label $y_i \in \mathcal{Y}$. Target test labels are strictly withheld from inference.

Pure episodic memory retains raw records $\mathcal{M}_{\text{raw}} = \mathcal{E}$ directly. Conversely, global semantic consolidation [18] compresses $\mathcal{E}$ into a global hypothesis set $\mathcal{H}^*$:

$$\mathcal{H}^* = \arg\max_{\mathcal{H}} P(\mathcal{H} \mid e_1, e_2, \ldots, e_n). \tag{1}$$

To preserve instance-level evidence while providing inspectable local heuristics, VLEG constructs a bank of source-verified episodic–semantic tuples (Figure 1):

$$\mathcal{M}_{\text{VLEG}} = \left\{ (e_i, r_i^*) \;\middle|\; e_i \in \mathcal{E},\ \mathcal{V}_\phi(e_i, r_i^*)_v = 1 \right\}, \tag{2}$$

where r_i^* is a natural-language detection rule induced from e_i, and $\mathcal{V}_\phi(e_i, r_i^*)_v \in \{0, 1\}$ is the binary acceptance decision of the source verifier.

3.2. Generator, Strict Source Verifier, and Bounded Cyclic Refinement

VLEG separates consolidation into two prompted roles executed via separate calls to the same model: the Reflection Generator $\mathcal{G}_\theta$ and the Strict Verifier $\mathcal{V}_\phi$ (Figure 2).

At initial attempt $j = 0$, the generator proposes a candidate reflection conditioned only on the label-free attributes x_i:

$$r_i^{(0)} \sim \mathcal{G}_\theta(\cdot \mid x_i). \tag{3}$$

In subsequent iterations ($1 \le j < K$), if the candidate is rejected, the generator receives x_i, its previous proposal $r_i^{(j-1)}$, and the verifier's critique $c_i^{(j-1)}$:

$$r_i^{(j)} \sim \mathcal{G}_\theta(\cdot \mid x_i, r_i^{(j-1)}, c_i^{(j-1)}). \tag{4}$$

The verifier checks $r_i^{(j)}$ against the full labeled source episode $e_i = (x_i, y_i)$ under a fail-closed native-Boolean JSON schema, outputting binary validity decision $v_i^{(j)} \in \{0, 1\}$ and critique $c_i^{(j)}$:

$$(v_i^{(j)}, c_i^{(j)}) = \mathcal{V}_\phi\big((x_i, y_i),\ r_i^{(j)}\big). \tag{5}$$

Let $j_i^* = \min\{0 \le j < K \mid v_i^{(j)} = 1\}$ denote the stopping iteration (Figure 3). If j_i^* exists, the loop commits $r_i^* = r_i^{(j_i^*)}$ to the Hybrid Memory Bank:

$$\mathcal{M} = \bigcup_{i=1}^{n} \left\{ (e_i, r_i^*) \;\middle|\; \exists\, 0 \le j < K \text{ s.t. } v_i^{(j)} = 1 \right\}. \tag{6}$$

3.3. Deterministic Counterexample Compiler and Cross-Episode Gating (VLEG-CG)

By construction, Equation (5) conditions $\mathcal{V}_\phi$ only on the single source episode $e_i = (x_i, y_i)$. While this prevents label inversions and fabricated features with respect to e_i, it has zero access to the remaining historical episodes $\mathcal{E}_{-i} = \mathcal{E} \setminus \{e_i\}$.

To bridge single-episode verification and cross-episode specificity without additional LLM calls, we introduce a deterministic counterexample compiler Ψ. Given a natural-language reflection r_i^* formatted as

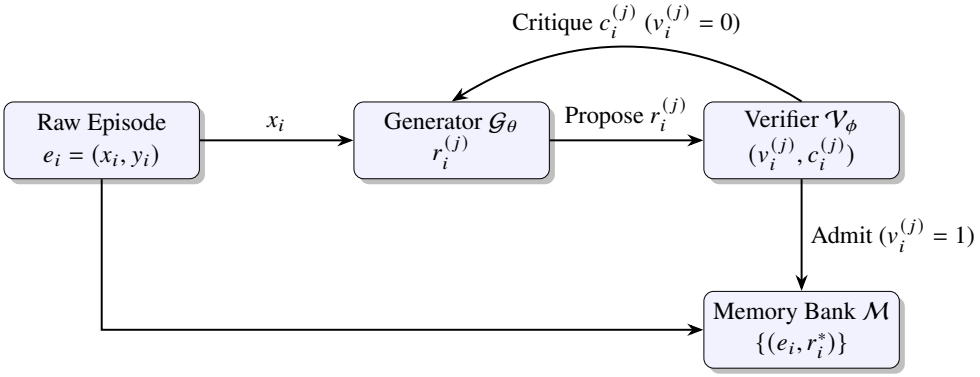


Figure 2. Architectural flow of VLEG memory consolidation.

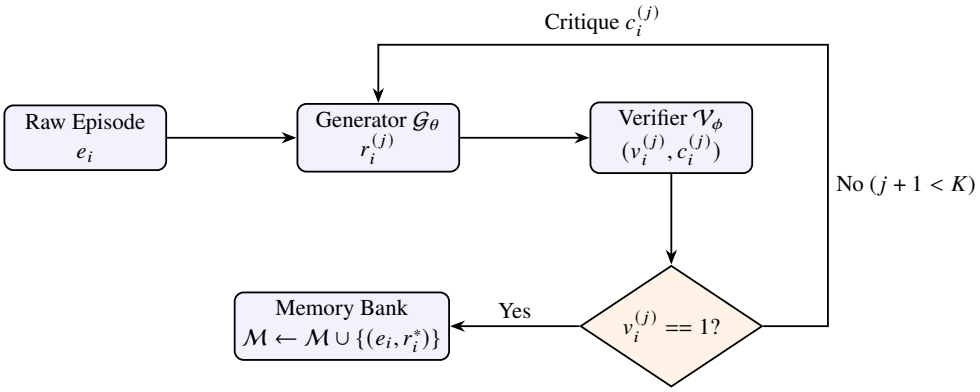


Figure 3. State transition diagram of bounded Iterative Cyclic Refinement.

a conjunctive `IF ... THEN ...` statement over flow attributes, Ψ parses the antecedent into a list of atomic feature comparisons $\pi(r_i^*)(x) = \bigwedge_{m=1}^{d_i} \mathrm{op}_m(x[\mathrm{field}_m], \mathrm{val}_m) \in \{0, 1\}$ and extracts the consequent class $\hat{y}(r_i^*) \in \{\texttt{BENIGN}, \texttt{MALICIOUS}\}$.

Definition 1 (Within-Bank Counterexample Count and Gate)**.** For an admitted tuple $(e_i, r_i^*) \in \mathcal{M}$, let $\mathrm{CE}(r_i^* \mid \mathcal{E}_{-i})$ denote the number of contrary-label episodes in the historical support bank $\mathcal{E}_{-i}$ that satisfy the compiled antecedent $\pi(r_i^*)$:

$$\mathrm{CE}(r_i^* \mid \mathcal{E}_{-i}) = \sum_{e_m = (x_m, y_m) \in \mathcal{E}_{-i}} \mathbb{I}\big(\pi(r_i^*)(x_m) = 1 \ \wedge \ y_m \neq \hat{y}(r_i^*)\big)\,. \tag{7}$$

We define two deterministic counterexample-gating policies over retrieved top-k tuples $\mathcal{M}_{\mathrm{top}}(T)$:

1. **Malicious-Rule Counterexample Gate (VLEG-CG-Mal):** Because false positives in intrusion triage are driven by overly broad malicious heuristics firing on benign flows, VLEG-CG-Mal flags a retrieved tuple (e_i, r_i^*) as over-generalized if $y_i = \texttt{MALICIOUS}$ and $\mathrm{CE}(r_i^* \mid \mathcal{E}_{-i}) > 0$ (or if r_i^* fails compilation). If any retrieved tuple in $\mathcal{M}_{\mathrm{top}}(T)$ is flagged, inference strips the verbal reflections and falls back to the raw admitted flow records $\{e_i\}_{(e_i, r_i^*) \in \mathcal{M}_{\mathrm{top}}(T)}$.
2. **All-Rule Counterexample Gate (VLEG-CG):** VLEG-CG applies the zero-counterexample constraint symmetrically across both classes, falling back to raw admitted flow records whenever any retrieved tuple in $\mathcal{M}_{\mathrm{top}}(T)$ has $\mathrm{CE}(r_i^* \mid \mathcal{E}_{-i}) > 0$.

3.4. Hybrid Episodic–Semantic Inference and Complete Algorithm

Given a target query T, VLEG retrieves the top-k memory tuples from $\mathcal{M}$. Let $z_i \in \{r_i^*, x_i\}$ denote the active indexing key for tuple $\mathcal{M}_i = (e_i, r_i^*)$, corresponding to reflection-embedding retrieval in symbolic tasks ($z_i = r_i^*$) and standardized feature-embedding retrieval in network flow tasks ($z_i = x_i$). Using dense embedding function

Algorithm 1 Verified Localized Episodic Grounding with Counterexample Gating (VLEG / VLEG-CG)

Input: Support episodes $\mathcal{E}$, target query T, budget K, retrieval count k, retriever R, gate mode Gate $\in$ {None, CG-Mal, CG-All}

Initialize Memory Bank $\mathcal{M} \leftarrow \emptyset$
// Phase 1: Bounded Source-Verified Consolidation
for each $e_i = (x_i, y_i) \in \mathcal{E}$ **do**
  $j \leftarrow 0$; $(r, c) \leftarrow (\emptyset, \emptyset)$; $v \leftarrow 0$
  while $v = 0$ **and** $j < K$ **do**
    $r_i^{(j)} \leftarrow \mathcal{G}_\theta(x_i, r, c)$ ▷ Generate or revise local reflection
    $(v, c) \leftarrow \mathcal{V}_\phi((x_i, y_i), r_i^{(j)})$ ▷ Verify against labeled source episode
    $r \leftarrow r_i^{(j)}$; $j \leftarrow j + 1$
  end while
  if $v = 1$ **then**
    $r_i^* \leftarrow r$; Compute $\mathrm{CE}(r_i^* \mid \mathcal{E} \setminus \{e_i\})$ via compiler Ψ (Equation (7))
    $\mathcal{M} \leftarrow \mathcal{M} \cup \{(e_i, r_i^*, \mathrm{CE}_i)\}$
  end if
end for
// Phase 2: Counterexample-Gated Hybrid Inference
$\mathcal{M}_{\mathrm{top}} \leftarrow R(\mathcal{M}, T, k)$
if Gate $\neq$ None **and** any $(e_i, r_i^*, \mathrm{CE}_i) \in \mathcal{M}_{\mathrm{top}}$ violates gate threshold **then**
  Context $\leftarrow$ PromptBuilder$(\{(e_i, \emptyset)\}_{(e_i, r_i^*) \in \mathcal{M}_{\mathrm{top}}}, T)$ ▷ Fallback to raw admitted flows
else
  Context $\leftarrow$ PromptBuilder$(\mathcal{M}_{\mathrm{top}}, T)$ ▷ Hybrid episode + rule context
end if
return I_ψ(Context)

$f_{\mathrm{emb}}(\cdot)$, similarity is computed as:

$$S(T, \mathcal{M}_i) = \frac{f_{\mathrm{emb}}(T) \cdot f_{\mathrm{emb}}(z_i)}{\|f_{\mathrm{emb}}(T)\| \|f_{\mathrm{emb}}(z_i)\|}, \qquad \mathcal{M}_{\mathrm{top}} = \underset{\mathcal{S} \subseteq \mathcal{M},\, |\mathcal{S}| = \min(k, |\mathcal{M}|)}{\arg\max} \sum_{(e_i, r_i^*) \in \mathcal{S}} S(T, (e_i, r_i^*)). \tag{8}$$

For standard VLEG (and un-gated queries under VLEG-CG), the inference prompt P pairs each retrieved episode with its verified reflection Q_i = [Episode: $e_i \oplus$ Verified Rule: r_i^*]; for counterexample-gated queries under VLEG-CG, Q_i = [Episode: e_i] retains only the verified source flow:

$$P = \mathrm{PromptBuilder}\left(\bigoplus_{(e_i, r_i^*) \in \mathcal{M}_{\mathrm{top}}} Q_i \oplus \text{Target: } T \right), \qquad \hat{y} = \mathrm{LLM}_\psi(P). \tag{9}$$

Algorithm 1 summarizes the unified VLEG and VLEG-CG pipeline.

4. Experiments and Diagnostic Results

4.1. Experimental Architecture

Our evaluation comprises two complementary stages:

1. **Controlled Symbolic Sanity Check (Sections 4.2–4.5):** We first use a deterministic 16-scene synthetic environment with a known ground-truth rule and controllable non-causal distractors (`texture` and `size`). Because this environment contains only 16 combinations ($n = 48$ pooled queries across 3 masking seeds at Level 3), we use it strictly as a controlled sanity check to verify how bounded refinement affects admission coverage, source-label agreement, and distractor mention rates under Qwen2.5-7B-Instruct and Llama-3.3-70B-Instruct (`nomic-embed-text`, temperature 0).

2. **Public Network Flow Benchmark, Counterexample Audit, and ML Baseline Comparison (Sections 4.6–4.7):** Our primary evaluation tests five historical-memory seeds (32 flows/seed: 16 benign, 16 malicious) over 1000 shared deduplicated UNSW-NB15 test fingerprints (663 benign, 337 malicious; 25,000 formal LLM predictions) on a dual-A800 server running Llama-3.3-70B-Instruct (`Q4_K_M`, $K = k = 3$, 32,768-token

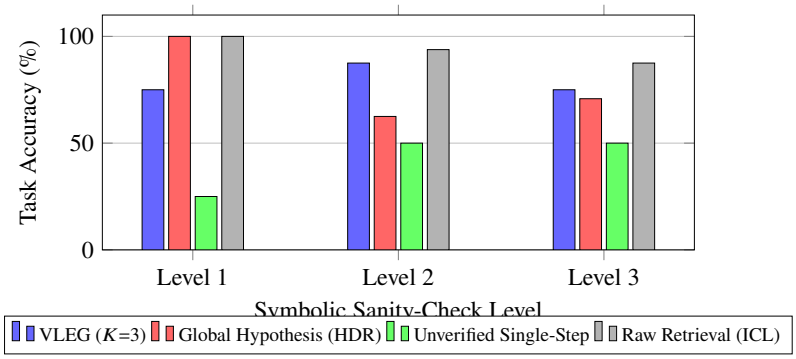


Figure 4. Symbolic sanity-check accuracy (%) across three complexity levels using Qwen2.5-7B-Instruct (Level 3 pooled across 3 masking seeds, $n = 48$).

context). We evaluate VLEG, VLEG-CG-Mal, VLEG-CG, raw retrieval, unverified reflection, admission-matched raw control, global summary, and both 32-flow few-shot and 88,699-flow full-train Random Forest and GBDT baselines.

4.2. Symbolic Sanity Check: Consolidation Strategies

On the 16-scene symbolic sanity check, VLEG eliminates the label-inversion collapse of unverified reflection and performs comparably to raw retrieval at 70B (44/48 **versus** 43/48**), while trailing raw retrieval at 7B (**75.0% **versus** 87.5%**).** In this synthetic environment, white backgrounds require a red circle for `ANOMALY`, black backgrounds invert this rule, and texture and size are non-causal distractors. Level 1 tests 8 white-background scenes; Level 2 tests all 16 scenes; and Level 3 withholds 4 scenes from the 12-scene support bank and evaluates all 16 scenes across three masking seeds ($n = 48$).

At 7B (Figure 4), raw retrieval (ICL) leads across Levels 1–3 (100.0%, 93.8%, and 87.5%) compared with VLEG (75.0%, 87.5%, and 75.0%). Global hypothesis consolidation (HDR adaptation) solves the uniform Level-1 rule (100.0%) but drops to 62.5% and 70.8% when background conditioning is required. Unverified single-step reflection collapses to the anomaly base rate (25.0%, 50.0%, and 50.0%) because unchecked proposals predominantly default to `ANOMALY`. Under Llama-3.3-70B-Instruct at Level 3 ($n = 48$), VLEG reaches 91.67% (44/48), which is comparable to raw retrieval at 89.58% (43/48, a one-query difference across 48 evaluations, with VLEG higher in two seeds and lower in one) and above global hypothesis consolidation (83.33%, 40/48) and binary-feedback Reflexion (81.25%, 39/48).

4.3. Symbolic Sanity Check: Grounding Modality Ablation

Under shared retrieval at 7B, hybrid grounding improves seen-combination accuracy (87.5%**) but trails single-modality context on withheld combinations (**60.0% **versus** 70.0%**).** Figure 5 compares Hybrid (e_i, r_i^*), Abstract-Only (r_i^*), and Raw-Only (e_i) across five Level-3 masking seeds ($n = 80$ overall, $n = 20$ masked-only). At 7B, Hybrid leads at Level 2 (87.5% versus 81.2%) but scores 73.8% overall and 60.0% on masked combinations at Level 3 (versus 78.8%/70.0% for Abstract-Only and 75.0%/70.0% for Raw-Only). Under Llama-3.3-70B-Instruct, Hybrid and Raw-Only tie at 95.0% overall (versus 91.25% for Abstract-Only), while Abstract-Only scores 85.0% on the 20 masked queries versus 80.0% for Hybrid and Raw-Only.

4.4. Symbolic Sanity Check: Refinement Budget and Overhead

Raising the refinement budget from $K = 1$ to $K = 3$ improves accuracy by recovering rejected episodes (83%–100% **versus** 42%–58% **coverage), with concave token growth.** Across five 7B masking seeds, $K = 3$ raises overall Level-3 accuracy from 63.75% (51/80) to 75.0% (60/80) and masked-only accuracy from 50.0% (10/20) to 60.0% (12/20). At 70B, $K = 3$ improves mean accuracy by 26.2 ± 8.1 pp over $K = 1$ and admits 100% (12/12) of historical episodes. Sweeping $K \in \{1, 2, 3, 5\}$ (Figure 6) shows concave token overhead at Level 2 (497, 776, 847, and 882 tokens/episode) because most episodes converge within two attempts.

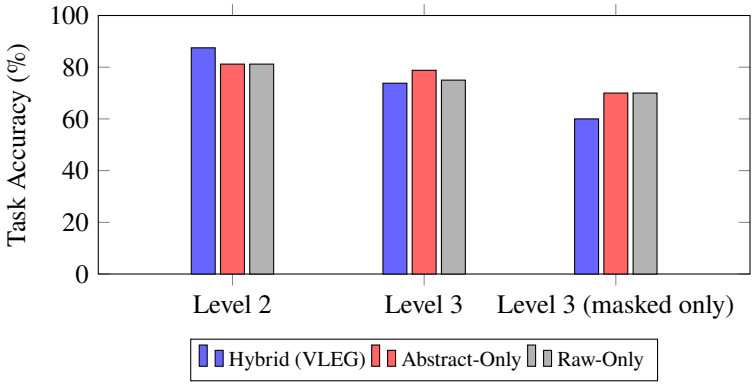


Figure 5. Grounding modality ablation under Qwen2.5-7B-Instruct (Level 3 pooled across 5 masking seeds, $n = 80$ overall, $n = 20$ masked-only).

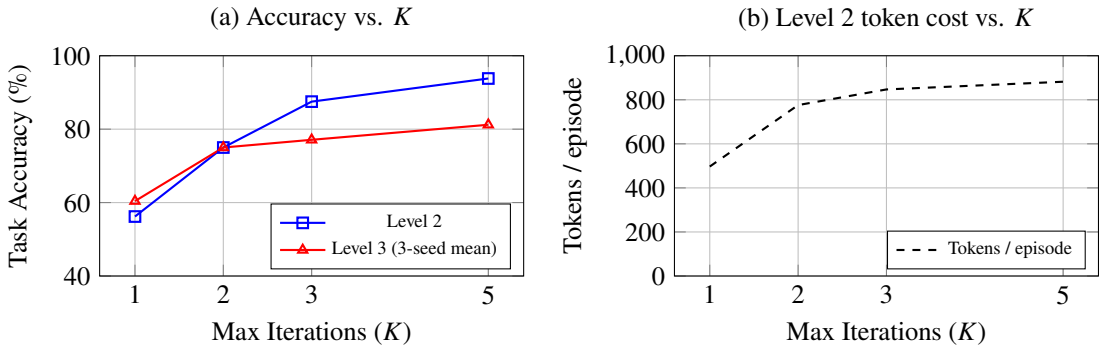


Figure 6. Sensitivity to refinement ceiling $K \in \{1, 2, 3, 5\}$ (Qwen2.5-7B-Instruct).

Table 1. Deterministic memory-quality diagnostics on the symbolic sanity check (Qwen2.5-7B-Instruct).

Level	Strategy	Admission Rate	Source-Label Agreement	Spurious (of correct)
Level 2	Unverified Single-Step	N/A (Always commit)	50.0% (8/16)	87.5% (7/8)
	VLEG (K=3)	**93.8% (15/16)**	**100.0% (15/15)**	86.7% (13/15)
Level 3	Unverified Single-Step	N/A (Always commit)	41.7% (5/12)	80.0% (4/5)
	VLEG (K=3)	**100.0% (12/12)**	**100.0% (12/12)**	83.3% (10/12)

4.5. Symbolic Sanity Check: Memory Quality Diagnostics

Single-episode verification achieves 100% **observed source-label agreement on admitted memories, but cannot filter out coincidental distractor attributes.** Table 1 shows that while VLEG raises source-label agreement from 41.7%–50.0% to 100.0% at 7B, 83.3%–86.7% of label-correct 7B reflections still cite non-causal texture or size attributes (versus 80.0%–87.5% for unverified proposals). Under Llama-3.3-70B-Instruct, distractor mention rates drop to 12.5% (2/16) at Level 2 and 16.7% (2/12) at Level 3 for both VLEG and unverified proposals alike, confirming that single-instance checks validate label consistency but not feature causality.

4.6. Public-Flow Benchmark on UNSW-NB15: Counterexample Audit and Gating

On deduplicated UNSW-NB15 flows, standard VLEG resolves the recall collapse of unverified reflection (+17.02 **pp Macro-F1) but suffers a** 48.78% **FPR due to over-generalized local rules; integrating our counterexample gate (VLEG-CG) slashes FPR by** 9.35–10.92 **pp and recovers Macro-F1 to** 63.32%–63.55%**.** We evaluate UNSW-NB15 flows [37] after 19-feature deduplication, conflicting-label exclusion, and strict train/test separation. Each of the five seeds samples 32 historical flows (16 benign, 16 malicious; 160 total historical episodes across seeds) and evaluates 1000 shared test fingerprints (663 benign, 337 malicious).

Tables 2–4 and Figure 7 establish four principal findings:

Table 2. Public-flow classification performance on the 1000 deduplicated UNSW-NB15 test fingerprints (mean ± sample SD across 5 memory seeds, in percent). All few-shot methods use the exact same 32 historical flows per seed (16 benign, 16 malicious). Full-train supervised baselines are trained on all 88,699 deduplicated training flows.

Supervision Regime	Method / Condition	Accuracy (%)	Macro-F1 (%)	Attack Recall (%)	False Positive Rate (%)
Few-Shot LLM Memory (32 flows/seed, k=3, Llama-3.3-70B, 0-train)	Raw flow retrieval	**65.08 ± 2.93**	**63.92 ± 2.54**	71.39 ± 8.44	38.13 ± 6.85
	Raw on admitted flows (control)	64.86 ± 2.96	63.70 ± 2.68	71.16 ± 8.71	38.34 ± 6.61
	Unverified single-step reflection	61.38 ± 1.08	42.66 ± 1.35	6.17 ± 1.43	**9.89 ± 2.03**
	Global summary	50.22 ± 12.45	47.60 ± 14.13	**87.12 ± 22.27**	68.54 ± 24.20
	VLEG (source-verified only, K=3)	59.74 ± 2.03	59.68 ± 2.03	77.74 ± 5.69	48.78 ± 1.67
	VLEG-CG-Mal (Ours, malicious CE gate)	**64.16 ± 1.92**	**63.32 ± 2.10**	**72.11 ± 8.20**	**39.43 ± 4.68**
	VLEG-CG (Ours, all-rule CE gate)	**64.66 ± 2.83**	**63.55 ± 2.49**	70.27 ± 8.14	**37.86 ± 7.16**
Few-Shot Tabular ML (Same 32 flows/seed)	Random Forest (32-flow few-shot)	68.86 ± 1.25	68.42 ± 1.23	87.60 ± 10.86	40.66 ± 5.66
	Gradient Boosting / GBDT (32-flow)	66.60 ± 4.46	65.84 ± 4.60	78.04 ± 10.27	39.22 ± 5.52
Full-Train Tabular ML (88,699 unique flows)	Random Forest (Full-train upper bound)	82.00	81.27	92.28	23.23
	Gradient Boosting / GBDT (Full-train)	80.50	79.88	93.47	26.09

Table 3. Deterministic counterexample compiler audit of generated traffic reflections across the 5 historical-memory seeds (160 total historical episodes: 80 benign, 80 malicious).

Method	Admitted	Benign Label Match	Malicious Label Match	Parseable Rules	Malicious Rules w/ CE	Mean CEs (Malicious)
Unverified single-step	160 / 160 (100.0%)	80 / 80 (100.0%)	**4 / 80 (5.0%)**	160 / 160 (100.0%)	73 / 80 (91.3%)	39.5
VLEG (K=3)	**158 / 160 (98.8%)**	**80 / 80 (100.0%)**	**78 / 78 (100.0%)**	**157 / 158 (99.4%)**	**50 / 77 (64.9%)**	**8.2**

Table 4. Observed public-flow memory construction and query costs (means across 5 memory seeds, 32 historical flows per seed). VLEG-CG and VLEG-CG-Mal execute deterministic Python rule screening during consolidation with zero additional LLM calls or tokens.

Condition	Construction Calls	Construction Tokens	Construction (s)	Query Mean (s)	Query p95 (s)
Raw flow retrieval	0	0	0	2.12	2.67
Unverified single-step	32.0	9405	126.54	2.23	2.75
VLEG / VLEG-CG (K=3)	120.4	40,094	419.43	1.74	2.68
Raw on admitted flows	120.4	40,094	419.43	0.95	1.19
Global summary	1.0	5947	23.21	1.21	1.30

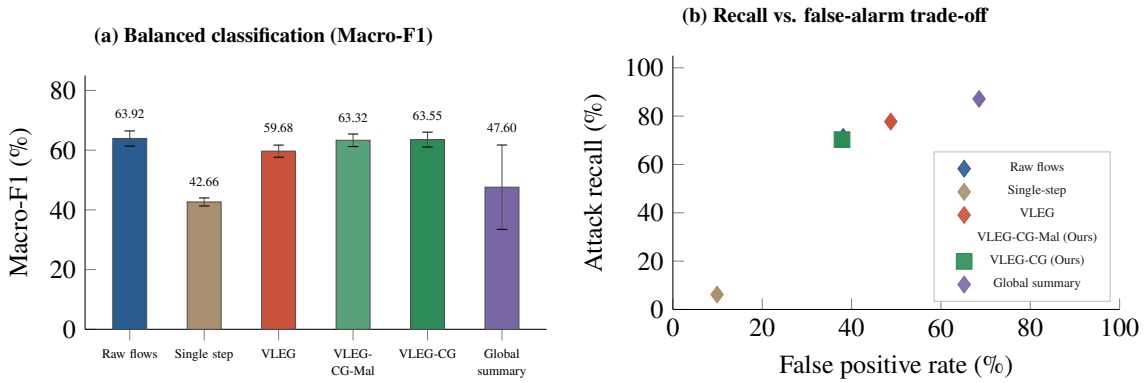


Figure 7. Public-flow evaluation across five historical-memory seeds on UNSW-NB15. (a) Mean Macro-F1 with sample SD bars; (b) attack recall versus false positive rate, showing how VLEG-CG-Mal and VLEG-CG suppress the false-alarm inflation of unscreened VLEG.

1. **Exact Cause of Unverified Reflection Collapse:** Table 3 reveals why unverified single-step reflection suffers a catastrophic attack-recall collapse (6.17% ± 1.43% recall, 42.66% Macro-F1, barely above an always-benign classifier at 39.87% Macro-F1). When generating rules from label-free flow attributes, the LLM correctly labels 80/80 (100%) benign flows as `BENIGN`, but **mislabels 76/80 (95.0%) of historical malicious flows as `BENIGN`** (identifying only 4/80 as `MALICIOUS`). Injecting these 97.5%-benign rules into inference causes the classifier to overwhelmingly predict benign on test attacks.

2. **Effect and Limitation of Single-Episode Verification:** VLEG's source verifier corrects this label halluci-

nation, achieving 100% source-label agreement across all 158/160 admitted episodes (80/80 benign, 78/78 malicious) and reducing the mean number of historical counterexamples per malicious rule from 39.5 to 8.2 (Table 3). Consequently, VLEG boosts Macro-F1 by +17.02 pp over single-step reflection (59.68%, paired 95% CI [13.71, 20.39] pp) and lifts attack recall to 77.74%. However, standard VLEG still trails zero-cost Raw flow retrieval (63.92% Macro-F1) by −4.24 pp (paired 95% CI [−5.91, −2.53] pp) because its FPR rises from 38.13% to 48.78%. Since the admission-matched control (`Raw on admitted flows`, 63.70% Macro-F1, 38.34% FPR) matches full Raw retrieval, the +10.65 pp FPR inflation is caused directly by the 50/77 (64.9%) admitted malicious rules and 63/80 (78.8%) benign rules that match contrary-label historical flows (113/157 = 72.0% overall).

3. **Effectiveness of Counterexample Gating (VLEG-CG-Mal and VLEG-CG):** When we apply our deterministic counterexample compiler strictly within each seed's 32-flow support bank ($\mathrm{CE}(r_i^* \mid \mathcal{E}_{\text{seed}} \setminus \{e_i\}) > 0$), **VLEG-CG-Mal** gates out over-generalized malicious rules on 590.2/1000 queries while retaining verified zero-counterexample malicious heuristics on the remaining 409.8/1000 queries. This slashes the FPR by **−9.35 pp** (from 48.78% to 39.43% ± 4.68%) and raises Macro-F1 by **+3.64 pp** (to 63.32% ± 2.10%), while preserving a higher attack recall (72.11% ± 8.20%) than both Raw flow retrieval (71.39%) and Raw on admitted flows (71.16%). Applying symmetric gating across both classes (**VLEG-CG**) reduces the FPR by **−10.92 pp** (to **37.86**% ± **7.16**%, the lowest FPR among all high-recall methods) and recovers Macro-F1 to **63.55**% ± **2.49**%.

4.7. Comparison with Traditional Supervised ML (RF, GBDT) and LLM-IDS Baselines

Evaluating Random Forest and GBDT on the exact same deduplicated UNSW-NB15 splits demonstrates that strict 19-feature deduplication makes the decision boundary intrinsically challenging (23.23%**–**26.09% **FPR even for full-train ensembles, and** 39.22%**–**40.66% **FPR in the 32-flow few-shot regime).** To address how LLM memory compares with conventional machine learning, Table 2 reports Random Forest (200 trees) and GBDT (100 trees) evaluated on the exact same 1000 test fingerprints under two supervision regimes:

- **Why Full-Train Accuracy is** 82.00% **Rather Than** 90%+ **(Deduplication Effect):** In raw UNSW-NB15 (175,341 training rows), tens of thousands of identical burst flows (e.g., repeated DNS/HTTP floods sharing identical 19-feature vectors) inflate raw test accuracy above 90%. Our benchmark enforces strict 19-feature deduplication, conflict removal, and train/test fingerprint disjointness, collapsing the training set to 88,699 unique flow fingerprints. On the resulting 1000 unique test fingerprints, full-train Random Forest achieves 82.00% accuracy (81.27% Macro-F1, 92.28% recall, 23.23% FPR) and full-train GBDT achieves 80.50% accuracy (79.88% Macro-F1, 93.47% recall, 26.09% FPR).
- **Same 32-Flow Few-Shot Regime:** When restricted to the exact same 32 historical flows per seed as the LLM methods, 32-flow Random Forest achieves 68.42% ± 1.23% Macro-F1 with a **40.66**% ± **5.66**% **FPR**, and 32-flow GBDT achieves 65.84% ± 4.60% Macro-F1 (78.04% recall) with a **39.22**% ± **5.52**% **FPR**. Notably, VLEG-CG (37.86% FPR) and VLEG-CG-Mal (39.43% FPR) achieve false positive rates on par with or slightly below 32-flow RF and GBDT, while generating human-auditable natural-language rule rationales alongside raw flow evidence—a capability absent in black-box tree ensembles and unscreened RAG IDS frameworks [5,6].

4.8. Methodological Limitations

Several boundaries constrain our findings. First, the 16-scene symbolic sanity check is a small synthetic environment where queries recur across masking seeds, and the 7B (Qwen2.5) and 70B (Llama-3.3) evaluations change model family, precluding causal isolation of parameter scale. Second, our UNSW-NB15 benchmark evaluates a 32-flow few-shot regime over deduplicated 19-feature vectors using a single quantized model (`Q4_K_M`); it does not represent line-rate packet inspection. Third, query latencies in Table 4 are confounded by fixed method execution order and KV-cache reuse, and 24 of 4968 identical-prompt pairs between Raw retrieval and Raw-on-admitted differed at temperature 0 due to floating-point non-determinism in quantized GPU kernels.

5. Conclusions

This paper presented Verified Localized Episodic Grounding (VLEG) and its counterexample-gated extension (VLEG-CG) to audit and mitigate rule over-generalization in few-shot LLM-assisted network traffic analysis. Across a 16-scene symbolic sanity check and a 25,000-prediction benchmark on 1000 deduplicated UNSW-NB15 test fingerprints, our investigation yields three core conclusions:

1. **Single-Episode Verification Prevents Label Inversion:** Unverified single-step reflection mislabels 95.0% (76/80) of malicious historical flows as benign, causing test attack recall to collapse to 6.17% (42.66% Macro-F1). Bounded per-episode verification ($K = 3$) restores 100% source-label agreement on admitted episodes (158/158) and raises attack recall to 77.74% (59.68% Macro-F1), while achieving accuracy comparable to raw retrieval on the 70B symbolic sanity check (44/48 versus 43/48).
2. **Single-Episode Fidelity Does Not Imply Cross-Flow Specificity:** Despite 100% source-label agreement and 40,094 construction tokens per seed, unscreened VLEG trails zero-cost raw flow retrieval (63.92% Macro-F1) by 4.24 pp because 72.0% (113/157) of compiled rules match contrary-label historical flows, inflating the false positive rate to 48.78%.
3. **Within-Bank Counterexample Gating Restores Precision:** Screening admitted rules with our deterministic counterexample compiler (VLEG-CG-Mal and VLEG-CG) at zero extra LLM cost cuts the false positive rate by 9.35–10.92 pp (to 37.86%–39.43%, matching 32-flow Random Forest and GBDT) and recovers Macro-F1 to 63.32%–63.55% while retaining verified verbal rules on un-gated queries.

In practical SOC deployments, full-train tabular classifiers (RF/GBDT achieving 80.50%–82.00% accuracy on deduplicated fingerprints) should serve as the primary high-throughput filter, whereas VLEG-CG provides an auditable second-stage triage assistant for cold-start or few-shot alert investigation. Future work will feed deterministic counterexample traces directly back into the generator prompt during Phase 1 refinement so that overly broad rules are rewritten rather than gated out at inference.

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